The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Canonical Image Restricted Projection / Canonical-Lift
Necessity / Uniqueness Theorem

Abstract

The preceding canonical image response / exact-shell necessity / uniqueness theorem already proves that once the fixed canonical image body $\mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}$ and the fixed restricted projection / canonical-lift relation on that body are held fixed, the canonical image response law $\mathcal{Q}_\sigma^{\mathrm{disp,resp,cbimg}}$ and the canonical exact-shell image $\mathcal{Z}_\sigma^{\mathrm{disp,resp,cbimg}}$ are uniquely forced. That spent the response-level and shell-level freedom inside the current image-core frontier, but it still left the relation itself as the live benchmark-facing burden on the fixed body.

The present note proves that this relation is forced as well by the remaining explicit projection / response data on the recorded route. More precisely, on the fixed canonical image body any benchmark-faithful lift section into that body whose ambient projection is the identity on the fixed current displacement body and whose induced quadratic response agrees with the already forced current body-response route must coincide fiberwise with the canonical minimal lift determined by the surjective displacement projection and the coercive positive self-adjoint substrate-response operator. Equivalently, the restricted projection on the fixed canonical image body is uniquely forced as the inverse of that lift. Therefore no genuinely different relation-level datum remains live inside the current image-core frontier once the body is fixed.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine sharper theorem strictly inside the current image-core frontier: the restricted projection / canonical-lift relation itself is no longer independently live there. What remains open is a genuinely cleaner deeper derivation of the full canonical current-body image core $\mathfrak{S}^{\mathrm{disp,resp,cbimg}}$, or a sharper theorem that makes that derivation unavoidable.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a stage where the honest next move is smaller than another deeper relay but sharper than another routing reminder. The preceding canonical image response / exact-shell necessity / uniqueness theorem already showed that once the fixed canonical image body and the fixed restricted projection / canonical-lift relation are in hand, the response-level and exact-shell constituents of the current image core carry no remaining benchmark-facing presentational freedom. After that gain, another manuscript that merely preserves the same response law and exact-shell image on the same body would no longer remove any live benchmark-facing freedom.

The next honest internal question is therefore one level narrower. Suppose the fixed canonical image body is already in hand. Does the restricted projection / canonical-lift relation on that body still remain independently live, or is that relation already forced by the remaining explicit projection / response data together with the fixed current body-response route? The present note proves the second statement. Any benchmark-faithful lift section from the fixed current displacement body into the fixed canonical image body that reproduces the already forced current body-response route must be the canonical minimal lift, and therefore the restricted projection on that body is forced as its inverse.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the already-screened larger substrate-body presentation, the larger projection-operator core, or the full displacement-response substrate law as if they were again the live burden. Its narrower benchmark-facing role is to prove that, once the canonical image body is already fixed, even the restricted projection / canonical-lift relation on that body is forced by the remaining explicit substrate projection / response data.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded proto-section benchmark base and current relation-forcing data)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes:*

1. *the current proto-section benchmark base* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and exact proto-section shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*
2. *the current section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{ppprese c}}$$

   whose fiber

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}}(y) \equiv \{v \in V_\sigma^{\mathrm{ppprese c}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*
3. *a finite-dimensional Euclidean substrate carrier*

$$U_\sigma^{\mathrm{disp,resp}};$$

4. *a smooth family of surjective linear displacement projections*

$$P_\sigma^{\mathrm{disp,resp}}(y) : U_\sigma^{\mathrm{disp,resp}} \to V_\sigma^{\mathrm{ppprese c}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}};$$

5. *a smooth family of positive self-adjoint substrate-response operators*

$$K_\sigma^{\mathrm{disp,resp}}(y) : U_\sigma^{\mathrm{disp,resp}} \to U_\sigma^{\mathrm{disp,resp}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

   satisfying the uniform coercive bound

$$\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y) u \right\rangle \geq \mu_\sigma \|u\|^2 \qquad \text{for all } y \in \mathcal{B}_\sigma^{\mathrm{psec}}, \quad u \in U_\sigma^{\mathrm{disp,resp}} \tag{2}$$

   for some constant $\mu_\sigma > 0$*;*
6. *the fixed exact-shell/current-body compatibility relation*

$$\{y \in \mathcal{B}_\sigma^{\mathrm{psec}} : 0 \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}(y)\} = \mathcal{Z}_\sigma^{\mathrm{psec}};$$

7. *benchmark faithfulness, meaning preservation of* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Recorded downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already records that, once the fixed canonical image body together with a benchmark-faithful restricted projection / canonical-lift relation on that body are available, the canonical image response law* $\mathcal{Q}_\sigma^{\mathrm{disp,resp,cbimg}}$ *and the canonical exact-shell image* $\mathcal{Z}_\sigma^{\mathrm{disp,resp,cbimg}}$ *are uniquely forced, and the current observer-relevant pullback body-response core* $\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyresp}}$*, the current pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyop}}$*, the current pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,op}}$*, the current pullback-quadratic core* $\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb}}$*, the current section-centered residual-normal-form realization* $\mathfrak{R}^{\mathrm{sub,proto,prese c,res}}$*, the current benchmark-faithful pre-proto-section package* $\mathfrak{R}^{\mathrm{sub,proto,prese c}}$*, the current minimizer-section core* $\mathfrak{R}^{\mathrm{sub,proto,prese c,sec}}$*, and the previously recorded continuity-Hessian compressed core and benchmark one-metric observer package follow unchanged.*

3. Canonical Lift and Fixed Image Body

Proposition 1 (Positive Schur-response operator)**.** *For each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, define*

$$M_\sigma^{\mathrm{disp,resp}}(y) \equiv P_\sigma^{\mathrm{disp,resp}}(y) K_\sigma^{\mathrm{disp,resp}}(y)^{-1} P_\sigma^{\mathrm{disp,resp}}(y)^* : V_\sigma^{\mathrm{ppprese c}} \to V_\sigma^{\mathrm{ppprese c}},$$

where $(\cdot)^*$ denotes Euclidean adjoint with respect to the fixed inner products on $U_\sigma^{\mathrm{disp,resp}}$ and $V_\sigma^{\mathrm{ppprese c}}$. Then $M_\sigma^{\mathrm{disp,resp}}(y)$ is positive self-adjoint and invertible for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$.

Proof. Self-adjointness is immediate from the self-adjointness of $K_\sigma^{\mathrm{disp,resp}}(y)^{-1}$. For any $0 \neq w \in V_\sigma^{\mathrm{ppprese c}}$,

$$\left\langle w, M_\sigma^{\mathrm{disp,resp}}(y) w \right\rangle = \left\langle K_\sigma^{\mathrm{disp,resp}}(y)^{-1/2} P_\sigma^{\mathrm{disp,resp}}(y)^* w, K_\sigma^{\mathrm{disp,resp}}(y)^{-1/2} P_\sigma^{\mathrm{disp,resp}}(y)^* w \right\rangle > 0$$

because $P_\sigma^{\mathrm{disp,resp}}(y)$ is surjective and therefore $P_\sigma^{\mathrm{disp,resp}}(y)^*$ is injective. Hence $M_\sigma^{\mathrm{disp,resp}}(y)$ is positive definite and invertible. □

Definition 4 (Canonical lift and current body-response route)**.** *For each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, define the* canonical minimal lift

$$J_\sigma^{\mathrm{disp,resp,can}}(y) \equiv K_\sigma^{\mathrm{disp,resp}}(y)^{-1} P_\sigma^{\mathrm{disp,resp}}(y)^* M_\sigma^{\mathrm{disp,resp}}(y)^{-1} : V_\sigma^{\mathrm{ppprese c}} \to U_\sigma^{\mathrm{disp,resp}}. \tag{3}$$

Define the core-forced body response

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,projop}}(y, v) \equiv \frac{1}{2} \left\langle J_\sigma^{\mathrm{disp,resp,can}}(y) v, K_\sigma^{\mathrm{disp,resp}}(y) J_\sigma^{\mathrm{disp,resp,can}}(y) v \right\rangle \qquad \text{for } (y, v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}. \tag{4}$$

Proposition 2 (Canonical lift properties)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$,*

1. $$P_\sigma^{\mathrm{disp,resp}}(y) \circ J_\sigma^{\mathrm{disp,resp,can}}(y) = \mathrm{id}_{V_\sigma^{\mathrm{ppprese c}}};$$

2. $$\left\langle J_\sigma^{\mathrm{disp,resp,can}}(y) v, K_\sigma^{\mathrm{disp,resp}}(y) k \right\rangle = 0$$

   for every $v \in V_\sigma^{\mathrm{ppprese c}}$ and every $k \in N_\sigma^{\mathrm{disp,resp}}(y) \equiv \ker P_\sigma^{\mathrm{disp,resp}}(y)$;

3. *every $u \in U_\sigma^{\mathrm{disp,resp}}$ admits a unique decomposition*

   $$u = J_\sigma^{\mathrm{disp,resp,can}}(y)\big(P_\sigma^{\mathrm{disp,resp}}(y) u\big) + k, \qquad k \in N_\sigma^{\mathrm{disp,resp}}(y);$$

4. *$J_\sigma^{\mathrm{disp,resp,can}}(y)$ is linear in the displacement argument and smooth in y.*

Proof. Item (1) follows directly from Definition 4. For item (2), let $k \in N_\sigma^{\mathrm{disp,resp}}(y)$. Then

$$\left\langle J_\sigma^{\mathrm{disp,resp,can}}(y) v, K_\sigma^{\mathrm{disp,resp}}(y) k \right\rangle = \left\langle P_\sigma^{\mathrm{disp,resp}}(y)^* M_\sigma^{\mathrm{disp,resp}}(y)^{-1} v, k \right\rangle = \left\langle M_\sigma^{\mathrm{disp,resp}}(y)^{-1} v, P_\sigma^{\mathrm{disp,resp}}(y) k \right\rangle = 0.$$

Item (3) follows by setting

$$k \equiv u - J_\sigma^{\mathrm{disp,resp,can}}(y)\big(P_\sigma^{\mathrm{disp,resp}}(y) u\big)$$

and applying item (1). Uniqueness follows because item (1) fixes the non-kernel part. Linearity and smoothness are immediate from (3). □

Proposition 3 (Unique energy-minimizing lift)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $v \in V_\sigma^{\mathrm{pppresec}}$, the canonical lift $J_\sigma^{\mathrm{disp,resp,can}}(y)v$ is the unique minimizer of*

$$\min \left\{ \frac{1}{2} \left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u \right\rangle : u \in U_\sigma^{\mathrm{disp,resp}},\ P_\sigma^{\mathrm{disp,resp}}(y)u = v \right\}. \tag{5}$$

Proof. Let $u \in U_\sigma^{\mathrm{disp,resp}}$ satisfy $P_\sigma^{\mathrm{disp,resp}}(y)u = v$. By Proposition 2(3),

$$u = J_\sigma^{\mathrm{disp,resp,can}}(y)v + k, \qquad k \in N_\sigma^{\mathrm{disp,resp}}(y).$$

Using Proposition 2(2),

$$\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u \right\rangle = \left\langle J_\sigma^{\mathrm{disp,resp,can}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)J_\sigma^{\mathrm{disp,resp,can}}(y)v \right\rangle + \left\langle k, K_\sigma^{\mathrm{disp,resp}}(y)k \right\rangle.$$

The last term is nonnegative by positivity of $K_\sigma^{\mathrm{disp,resp}}(y)$, and by the coercive bound (2) it vanishes only if $k = 0$. Therefore the unique minimizer of (5) is $J_\sigma^{\mathrm{disp,resp,can}}(y)v$. □

Definition 5 (Canonical image body, restricted projection / lift relation, and canonical image response / shell)**.** *For each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, define the* canonical image body

$$\mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y) \equiv J_\sigma^{\mathrm{disp,resp,can}}(y)\big(\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)\big) \subset U_\sigma^{\mathrm{disp,resp}}. \tag{6}$$

Define the canonical image response law

$$\mathscr{Q}_\sigma^{\mathrm{disp,resp,cbimg}}(y,u) \equiv \frac{1}{2} \left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u \right\rangle \qquad \text{for } u \in \mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y), \tag{7}$$

and the canonical exact-shell image

$$\mathcal{Z}_\sigma^{\mathrm{disp,resp,cbimg}} \equiv \{(y,0) : y \in \mathcal{Z}_\sigma^{\mathrm{psec}}\}. \tag{8}$$

Proposition 4 (Restricted projection is inverse to the canonical lift on the image body)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, the restricted projection*

$$P_\sigma^{\mathrm{disp,resp}}(y)\big|_{\mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y)} : \mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y) \to \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$$

is a bijection with inverse $J_\sigma^{\mathrm{disp,resp,can}}(y)\big|_{\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)}$.

Proof. Surjectivity follows from the definition (6). If $u \in \mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y)$, then $u = J_\sigma^{\mathrm{disp,resp,can}}(y)v$ for some $v \in \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$, and Proposition 2(1) gives

$$P_\sigma^{\mathrm{disp,resp}}(y)u = v.$$

Hence the restricted projection is inverse to the canonical lift on the stated domains and is therefore bijective. □

Remark 1. *The preceding canonical-image-body necessity / uniqueness theorem already spent the*

body-level freedom inside the current image-core frontier, and the later canonical-image-response / exact-shell necessity / uniqueness theorem already spent the response-level and shell-level freedom once the relation is fixed. The present note asks whether even this restricted projection / canonical-lift relation still carries independent benchmark-facing freedom once the body is fixed.

4. Compatible Restricted Projection / Lift Packages on the Fixed Image Body

Definition 6 (Benchmark-faithful compatible current-image restricted projection / canonical-lift package)**.** *A* benchmark-faithful compatible current-image restricted projection / canonical-lift package *for sector σ consists of inverse sections*

$$\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y) : \mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \to \mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y), \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

such that:

1. *the ambient projection is identity on the fixed current displacement body:*

$$P_\sigma^{\mathrm{disp,resp}}(y)\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v = v$$

   for every $(y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}$;

2. *the induced quadratic response agrees with the fixed current body-response route:*

$$\frac{1}{2}\left\langle \widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v \right\rangle = \widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,projop}}(y, v)$$

   for every $(y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}$;

3. *zero displacement on the proto-section shell is represented by the zero section:*

$$\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)0 = 0 \qquad \text{for every } y \in \mathcal{Z}_\sigma^{\mathrm{psec}};$$

4. *benchmark faithfulness in the sense of Definition 2(7).*

Remark 2. *Definition 6 is intentionally internal to the already-fixed image-body frontier. It does not reopen the larger substrate-body presentation, the larger projection-operator core, the full displacement-response substrate law, or the already-spent response / exact-shell data. It asks only whether, once the fixed image body is held fixed, any genuinely different benchmark-facing relation-level datum can remain on that body.*

Proposition 5 (The canonical restricted projection / lift package is compatible)**.** *The canonical lift $J_\sigma^{\mathrm{disp,resp,can}}(y)|_{\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)}$ forms a benchmark-faithful compatible current-image restricted projection / canonical-lift package.*

Proof. Item (1) is Proposition 2(1). Item (2) is Definition 4. For item (3), linearity gives $J_\sigma^{\mathrm{disp,resp,can}}(y)0 = 0$. Item (4) holds because the fixed image body together with the same current body-response route preserve the same benchmark-facing downstream package by Definitions 2(7) and 3. □

Proposition 6 (Compatible lifts are canonical)**.** *Let* $\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}$ *be any benchmark-faithful compatible current-image restricted projection / canonical-lift package. Then*

$$\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v = J_\sigma^{\mathrm{disp,resp,can}}(y)v$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ *and every* $v \in \mathcal{D}_\sigma^{\mathrm{ppresec}}(y)$*.*

Proof. Fix $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and $v \in \mathcal{D}_\sigma^{\mathrm{ppresec}}(y)$. By Definition 6(1), the point $\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v$ satisfies

$$P_\sigma^{\mathrm{disp,resp}}(y)\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v = v.$$

Hence it is an admissible point in the constrained minimization problem (5). By Definition 6(2),

$$\frac{1}{2}\left\langle \widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v\right\rangle = \widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,projop}}(y,v) = \frac{1}{2}\left\langle J_\sigma^{\mathrm{disp,resp,can}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)J_\sigma^{\mathrm{d}}\right.$$

Therefore $\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v$ achieves the same minimal value as the canonical lift. Proposition 3 says that minimizer is unique. Hence $\widetilde{J}_\sigma^{\mathrm{disp,resp,cbimg}}(y)v = J_\sigma^{\mathrm{disp,resp,can}}(y)v$. □

Corollary 1 (The restricted projection / canonical-lift relation is forced)**.** *For every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, the restricted projection*

$$P_\sigma^{\mathrm{disp,resp}}(y)\big|_{\mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y)} : \mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}(y) \to \mathcal{D}_\sigma^{\mathrm{ppresec}}(y)$$

is a bijection with unique benchmark-faithful inverse

$$J_\sigma^{\mathrm{disp,resp,can}}(y)\big|_{\mathcal{D}_\sigma^{\mathrm{ppresec}}(y)}.$$

Proof. Proposition 4 gives the stated bijection and identifies one inverse, namely the canonical lift. Proposition 6 shows that every benchmark-faithful compatible inverse section coincides with that canonical lift. Hence the relation is uniquely forced at benchmark scope. □

Proposition 7 (The canonical relation is sufficient at current benchmark scope)**.** *The fixed canonical image body together with the canonical restricted projection / canonical-lift relation is sufficient to determine every benchmark-facing downstream statement currently available below the current image-core frontier.*

Proof. By Proposition 5, the canonical relation is a benchmark-faithful compatible current-image restricted projection / canonical-lift package. Definition 3 then gives the canonical image response law, the canonical exact-shell image, the current pullback body-response core $\mathfrak{S}^{\mathrm{sub,proto,ppresec,pb,bodyresp}}$, the current pullback body-operator core $\mathfrak{S}^{\mathrm{sub,proto,ppresec,pb,bodyop}}$, the current pullback-operator core $\mathfrak{S}^{\mathrm{sub,proto,ppresec,pb,op}}$, the current pullback-quadratic core $\mathfrak{S}^{\mathrm{sub,proto,ppresec,pb}}$, the current residual-normal-form realization $\mathfrak{R}^{\mathrm{sub,proto,presec,res}}$, the current benchmark-faithful pre-proto-section package $\mathfrak{R}^{\mathrm{sub,proto,presec}}$, the current minimizer-section core $\mathfrak{R}^{\mathrm{sub,proto,presec,sec}}$, and the previously recorded continuity-Hessian compressed core and benchmark one-metric observer package unchanged. □

5. Main Theorem

Theorem 1 (Canonical image restricted projection / canonical-lift necessity / uniqueness on the current route)**.** *Fix the primitive continuation data of Definition 1, the recorded proto-section benchmark base and current relation-forcing data of Definition 2, and the recorded downstream chain of Definition 3. Then, once the fixed canonical image body* $\mathcal{U}_\sigma^{\mathrm{disp,resp,cbimg}}$ *is held fixed, the restricted projection / canonical-lift relation on that body is the unique benchmark-facing relation compatible with the fixed current displacement body, the fixed surjective projection, the fixed coercive response operator, and the fixed current body-response route. More precisely:*

1. *every benchmark-faithful compatible current-image restricted projection / canonical-lift package has inverse section equal fiberwise to the canonical lift by Proposition 6;*
2. *consequently the ambient restricted projection on the fixed canonical image body is the unique benchmark-faithful restricted projection / canonical-lift relation by Corollary 1;*
3. *the fixed canonical image body together with the canonical relation is sufficient to determine every currently available benchmark-facing downstream consequence by Proposition 7;*
4. *therefore no genuinely different benchmark-facing relation-level datum remains live inside the current image-core frontier once the body is fixed.*

Proof. Item (1) is Proposition 6. Item (2) is Corollary 1. Item (3) is Proposition 7. Item (4) is the routing consequence of items (1)–(3): any alternative compatible lift is not genuinely alternative because it coincides with the canonical one, while any purported relation that changes the fixed current body-response route or the downstream benchmark package is not benchmark-faithfully compatible with the current route. □

Corollary 2 (No genuinely different relation datum remains at current scope)**.** *Inside the present section-centered hidden-response current-body-image-core frontier, there is no genuinely different benchmark-facing relation-level datum below the fixed canonical image body once the fixed current displacement body, the fixed surjective projection, the fixed coercive response operator, and the fixed current body-response route are held fixed.*

Proof. By Theorem 1, every benchmark-faithful compatible restricted projection / canonical-lift package coincides with the canonical one. Hence no distinct compatible relation-level datum remains live at current benchmark scope. □

Corollary 3 (Primary post-relation burden)**.** *After Theorem 1, the next honest benchmark-facing manuscript below the present frontier must do at least one of the following:*

1. *derive or force the full canonical current-body image core* $\mathfrak{S}^{\mathrm{disp,resp,cbimg}}$ *in one genuinely cleaner deeper package from still more primitive explicit substrate structure;*
2. *prove a sharper compatibility, necessity, or uniqueness theorem that makes such a full-image-core derivation unavoidable while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary;*
3. *do one of the previous two without reopening the larger substrate-body presentation, the larger projection-operator core, or the full displacement-response substrate law as if those already-spent larger presentations were again the live burden.*

Another same-output deeper relay that merely preserves the same canonical image body, the same restricted projection / canonical-lift relation, the same image response law, the same exact-shell image, and therefore the same full current image core, another restoration of the full substrate-body presentation, or another replay of the larger projection-operator or full displacement-response substrate-law presentations is not the honest default next move.

Proof. The preceding canonical-image-body necessity / uniqueness theorem already spent the body-level freedom inside the current image-core frontier, and the canonical-image-response / exact-shell necessity / uniqueness theorem already spent the response-level and shell-level freedom once the relation is fixed. Theorem 1 and Corollary 2 now show that the relation-level freedom is spent as well. Therefore a new primary manuscript must either derive the whole current image core in a genuinely cleaner deeper package or prove a theorem that forces such a derivation. If it does not, then it preserves the same already-forced image-core data and is benchmark neutral at project-status level. □

Proposition 8 (The benchmark package remains fixed)**.** *The present necessity / uniqueness theorem preserves the benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definitions 2(7) and 6(4). The present note removes only the remaining relation-level presentational freedom inside the current image-core frontier. □

6. Conclusion

The positive result of this note is narrower than a completed substrate derivation of GR but sharper than another routing reminder. On the current section-centered hidden-response route, once the canonical image body is fixed, the restricted projection / canonical-lift relation on that body is now proved to be uniquely forced by the fixed surjective projection, the fixed coercive response operator, and the fixed current body-response route.

That matters because it spends the remaining relation-level presentational freedom inside the current image-core frontier. Any benchmark-faithful compatible lift section into the fixed canonical image body is forced to be exactly the canonical minimal lift, and the ambient restricted projection on that body is forced as its inverse. The earlier canonical-image-body necessity / uniqueness theorem remains preserved main-line history because it first removed the body constituent as an independently live burden. The canonical-image-response / exact-shell necessity / uniqueness theorem remains preserved main-line history because it already shows that once the relation proved here is fixed, the remaining response-level and shell-level data are forced as well.

The scope remains honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. The next open burden is therefore no longer to restate or relocate the same image body, the same restricted projection / canonical-lift relation, the same image response law, or the same exact-shell image through another deeper relay. It is to derive or justify the full canonical current-body image core $\mathfrak{S}^{\mathrm{disp,resp,cbimg}}$ in a genuinely cleaner deeper package, or to prove a sharper theorem that makes that derivation unavoidable, while preserving the same benchmark one-metric package and the same claim boundary.